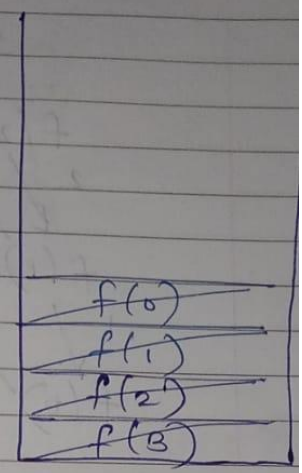
Assignment - 6

1)
```
psvm () {
    int n = 3
    fun (n);
}
void fun( int n) {
    if (n==0) {
        return;
    }
    Sop (n+" ");
    fun (n-1)
    Sop ( (n+1) + " ");
}
```

O/P
3
2
1
2
3
4

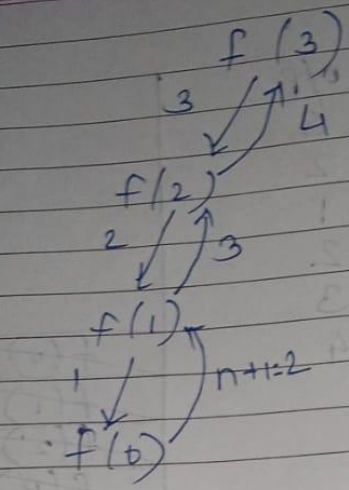



```
void fun (int n) {
    if (n==0)
        return
    Sop (n);
    fun (n-1)
    Sop (n+1);
```

```
void fun (int n) {
    if (n==0)
        return
    Sop (n)
    fun (n-1)
    Sop (n+1)
```

```
void fun (int n) {
    if (n==0)
        return
    Sop (n)
    fun (n-1)
    Sop (n+1)
```

recursive tree.



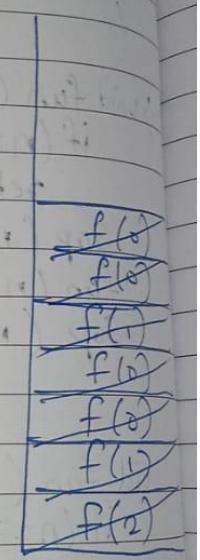
total calls = 4

2)

```
psvm ( ){
    int n = 2 ;
    fun ( n ) ;
}
                2
void fun ( int n ){
    if ( n == 0 )
        return ;
    Sop ( " x " ) ;
    fun ( n-1 )
    Sop ( " y " ) ;
    fun ( n-1 ) ;
}
```

o/p

x
x
y
y
x
y




```
                1
void f ( int n ){
    if ( n == 0 )
        return
    Sop ( " x " )
    f ( n-1 )
    Sop ( " y " )
    f ( n-1 )
}
```

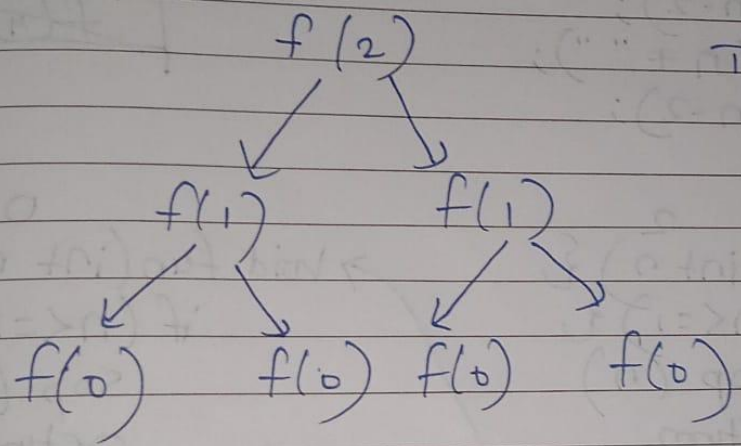
```
                0
void f ( int n ){
    if ( n == 0 )
        return
```

```
                0
void f ( int n ){
    if ( n == 0 )
        return
```

```
f(int n){
  if (n==0)
    return
  Sop ("x");
  f(n-1)
  Sop ("y");
  f(n-1)
```

```
f(int n){
  if (n==0)
    return
```

```
f(int n){
  if (n==0)
    return
```



Total Calls = 7

3]

```
psvm() {
    int n = 4;
    fun (n);
}
```

```
void fun (int n) {
    if (n<=1) {
        Sop (n + " ");
        return;
    }
    fun (n-2);
    Sop (n + " ");
    fun (n-2);
}
```

```
void fun (int n) {
    if (n<=1) {
        Sop (n)
        return
    }
    fun (n-2)
    Sop (n)
    fun (n-2)
}
```

```
void fun (int n) {
    if (n<=1) {
        Sop (n)
        return
    }
    fun (n-2)
    Sop (n)
    fun (n-2)
```

```
void fun (int n) {
    if (n<=1) {
        Sop (n)
        return;
    }
```

```
void fun (int n) {
    if (n<=1) {
        Sop (n)
        return;
    }
```

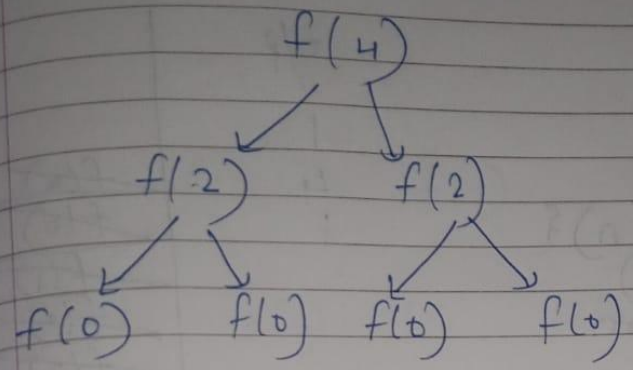
```
void fun (int n) {
    if (n<=1) {
        Sop(n)
        return
```

```
void fun (int n) { if (n<=
    Sop(n) return
```

o/p

0
2
0
4
0
2
0

~~f(0)~~
~~f(0)~~
~~f(2)~~
~~f(0)~~
~~f(0)~~
~~f(2)~~
~~f(4)~~



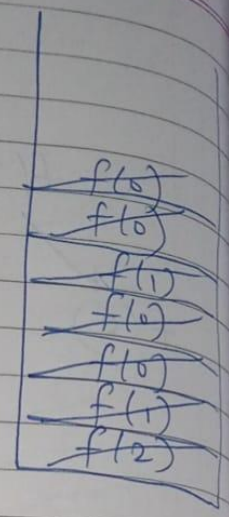
Calls = 7

4)

```
psvm(){
    int n=2;
    fun(n)
}
void fun(int n){
    if(n==0)
        return
    f(n-1)
    Sop(n*n)
    f(n-1)
}
```

o/p

1
4
1




```
void fun(int n){
    if(n==0)
        return
    f(n-1)
    Sop(n*n)
    f(n-1)
```

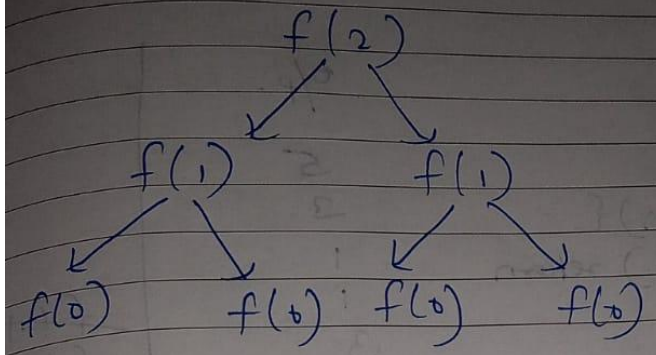
```
void fun(int n){
    if(n==0)
        return
```

```
void fun(int n){
    if(n==0)
        return
```

```
void fun(int n){
    if(n==0)
        return
    f(n-1)
    Sop(n*n)
    f(n-1)
```

```
void fun(int n){
    if(n==0)
        return
```

```
void fun(int n){
    if(n==0)
        return
```



Calls : 7

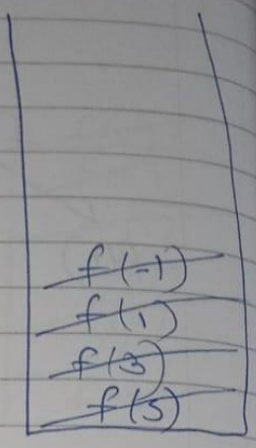
5)

```
psvm () {
  int n = 5;
  fun (n);
}
```

```
void fun (int n){
    if (n<=0) return
    Sop (n)
    fun (n-2)
    Sop (n)
```

o/p.

5
3
1
1
3
5




```
void fun(int n){
    if (n<=0) return
    Sop (n)
    fun (n-2)
    Sop (n)
```

```
void fun (int n){
    if (n<=0){
        return
    Sop (n)
    fun (n-2)
    Sop (n)
```

```
void fun (int n){
    if (n<=0) return
```

f(5)

f(3)

f(1)

f(-1)

Calls = 4

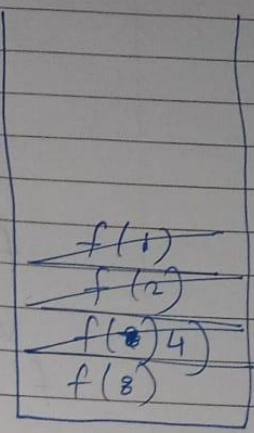
6)

```
psvm () {
  int n = 8
  fun (n);
}

                8
void fun (int n){
   if (n<=1){
        Sop (n)
        return
   }
   Sop (n)
   fun (n/2);
   Sop (n)
```

o/p -

8
4
2
1
2
4
8




```
                4
void fun (int n){
   if (n<=1){
        Sop (n)
        return
   }
   Sop (n)
   fun (n/2);
   Sop (n)
```

```
                2
void fun (int n){
   if (n<=1){
        Sop (n)
        return
   }
   Sop (n)
   fun (n/2)
   Sop (n)
```

```
                1
void fun (int n){
   if (n<=1){
        Sop (n)
        return
```

f(8)

↓

f(4)

↓

f(2)

↓

f(1)

Calls = 4

7]

```
psvm () {
int n = 3
fun(n);
}
```

```
3
void fun (int n){
    if (n<=0) return
    Sop ("A" + n);
    fun (n-1)
    Sop ("B" + (n*2));
```

o/p.

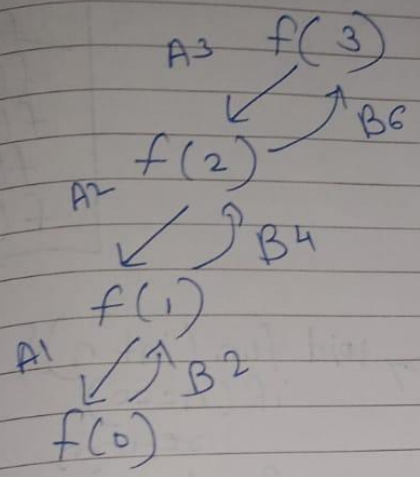
A 3
A 2
BA 1
B 2
B 4
B 6

~~f(0)~~
~~f(1)~~
~~f(2)~~
f(3)

```
2
void fun (int n)
    if (n<=0)
        return
    Sop (A + n)
    fun (n-1)
    Sop ("B" + n×2)
```

```
1
void fun (int n)
    if (n<=0)
        return
    sop (A + n)
    fun (n-1)
    Sop ("B" + n*2)
```

```
void fun (int n){
    if (n<=0)
        return
```



Calls = 4

```
psvm ( ){
  int n = 3
  fun (n)
}
void fun (int n)3 {
    if (n<=1){
        Sop (n)
        return
    }
    fun (n-1)
    Sop (n)
    fun (n-2)
}
```

```
void fun (int n)2 {
    if (n<=1) {
        Sop (n)
        return
    }
    fun (n-1)
    Sop (n)
    fun (n-2)
}
```

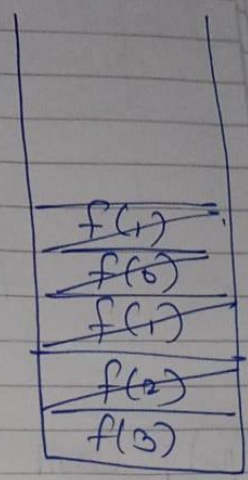
```
void fun (int n)1 {
    if (n<=1) {
        Sop (n)
        return
```

```
void fun (int n)1 {
    if (n<=1)
        Sop (n)
        return
```

```
void fun (int n)0 {
    if (n<=1)
        Sop (n)
        return
```

o/p -

1
2
0
3
1



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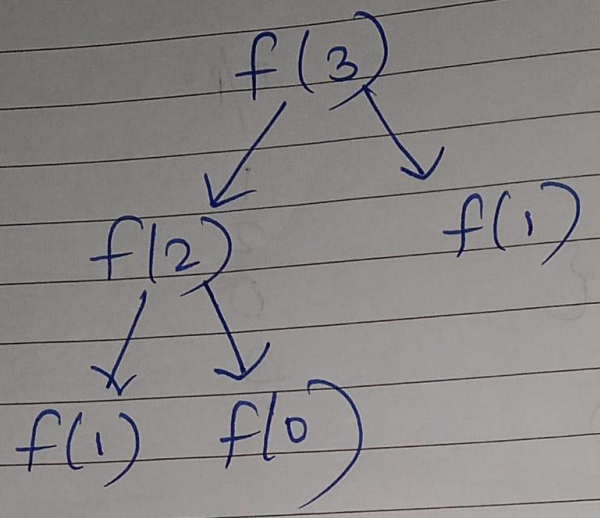
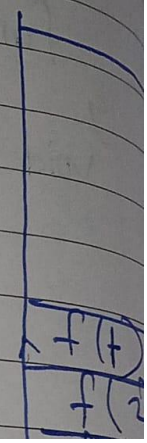
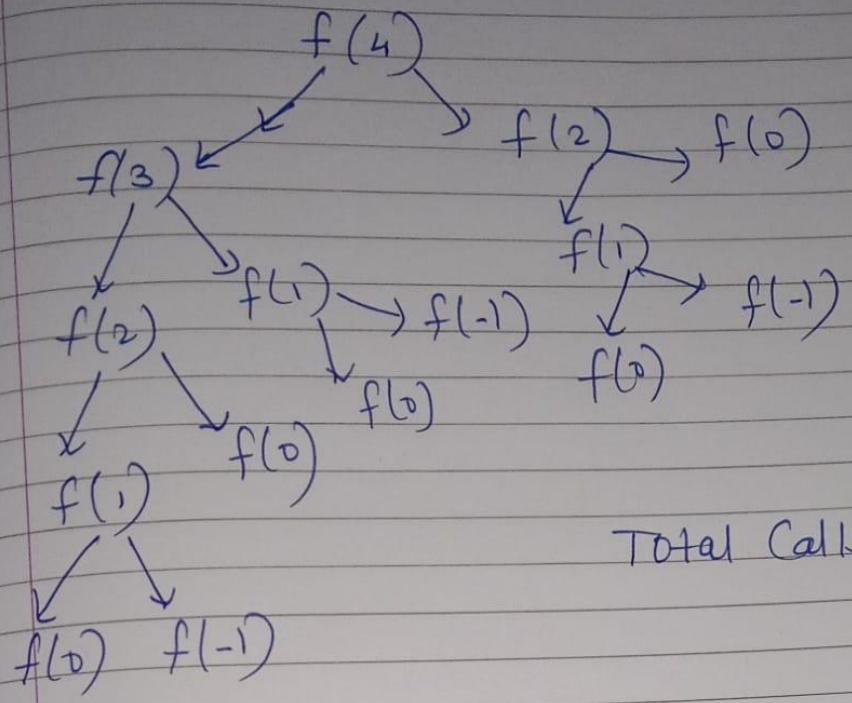


Calls = 5



```
psvm () {
  int n = 4;
  fun(n)
}

void fun (int n)[4] {
    if (n<=0) return
    f(n-1)
    f(n-2)
}

void fun (int n)[3] {
    if (n<=0) return
    f(n-1)
    f(n-2)

void fun (int n)[2] {
    if (n<=0) return
    f(n-1)
    f(n-2)

void fun (int n)[1] {
    if (n<=0) return
    f(n-1)
    f(n-2)

void fun (int n)[0] {
    if (n<=0) return

void fun (int n)[-1] {
    if (n<=0) return

void fun (int n)[0] {
    if (n<=0) return

void fun (int n)[-1] {
    if (n<=0) return

void fun (int n)[1] {
    if (n<=0) return
    f(n-1)
    f(n-2)

void fun (int n)[0] {
    if (n<=0) return

void fun (int n)[-1] {
    if (n<=0) return

void fun (int n)[0] {
    if (n<=0) return

void fun (int n)[-1] {
    if (n<=0) return
```

Q. 10 Count Calls



Total Calls = 15

~~f(0)~~
~~f(-1)~~
~~f(0)~~
~~f(1)~~
~~f(2)~~
~~f(-1)~~
~~f(0)~~
~~f(1)~~
~~f(0)~~
~~f(-1)~~
~~f(0)~~
~~f(1)~~
~~f(2)~~
~~f(3)~~
~~f(4)~~